

Confidence Intervals Using the t-Distribution

Statistics Worksheet · Grade 11–College

Name: _____

Date: _____

Learning Objectives

- Identify the point estimate, margin of error, standard error, and sample standard deviation from a given confidence interval
- Use the t-distribution formula to work backwards from a confidence interval to find unknown statistics
- Interpret a confidence interval in a statistically correct context

Problems

1. A 95% confidence interval for the mean lifetime of AA batteries is (430, 470) minutes. What is the point estimate (sample mean) for this interval?

(430, 470)

2. Using the confidence interval (430, 470) minutes, find the margin of error.

(430, 470)

3. A 90% confidence interval for a population mean is (118, 132). Find the point estimate and the margin of error.

(118, 132)

4. For a t-distribution confidence interval with a 95% confidence level and sample size $n = 30$, find the tail area used to look up the critical t-value.

$$\frac{1 - 0.95}{2}$$

5. For a 95% confidence interval with sample size $n = 30$, determine the degrees of freedom and find the critical t-value (t^*) using the inverse-t function.

$$df = n - 1$$



6. Given a margin of error of 20, a critical t-value of 2.045, and a sample size of 30, use the margin of error formula to find the sample standard deviation.

$$E = t^* \cdot \frac{s}{\sqrt{n}}$$

7. Using the margin of error of 20 and the critical t-value of approximately 2.045, find the standard error of the mean.

$$SE = \frac{E}{t^*}$$

8. Write a statistically correct interpretation of the 95% confidence interval (430, 470) minutes for the mean lifetime of AA batteries.

9. A 99% confidence interval for the mean weight of a cereal box is (15.6, 16.4) ounces with a sample size of 25. Find the point estimate, margin of error, critical t-value, and sample standard deviation.

(15.6, 16.4), $n = 25$, 99%

10. Two confidence intervals are constructed for the same population mean using the same sample: one at 90% confidence and one at 99% confidence, both with $n = 20$. The 90% interval is (54.3, 65.7). Without recalculating from raw data, determine the point estimate, find the 90% margin of error, find t^* for 90% and 99% ($df = 19$), calculate the standard deviation, and then construct the 99% confidence interval.

(54.3, 65.7), $n = 20$, 90% and 99%



Confidence Intervals Using the t-Distribution — Answer Key

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Answer Key

1. Answer: 450 minutes

- The point estimate is the midpoint of the confidence interval.
- Point estimate = $(430 + 470) / 2 = 900 / 2 = 450$

2. Answer: 20 minutes

- The margin of error is the distance from the midpoint to either endpoint.
- Point estimate = 450 (midpoint)
- Margin of error = $470 - 450 = 20$ (or equivalently $450 - 430 = 20$)

3. Answer: Point estimate = 125; Margin of error = 7

- Point estimate = $(118 + 132) / 2 = 250 / 2 = 125$
- Margin of error = $132 - 125 = 7$

4. Answer: 0.025

- Tail area = $(1 - \text{confidence level}) / 2$
- Tail area = $(1 - 0.95) / 2 = 0.05 / 2 = 0.025$

5. Answer: df = 29; $t^* \approx 2.045$

- Degrees of freedom = $n - 1 = 30 - 1 = 29$
- Tail area = 0.025 (from 95% confidence level)
- Using inverse-t: $t^*(0.025, 29) \approx -2.045$; use the positive value $t^* \approx 2.045$

6. Answer: $s \approx 53.57$

- $E = t^* \times (s / \sqrt{n}) \rightarrow 20 = 2.045 \times (s / \sqrt{30})$
- Multiply both sides by $\sqrt{30}$: $20\sqrt{30} = 2.045 \times s$
- $s = (20 \times \sqrt{30}) / 2.045 = (20 \times 5.477) / 2.045 \approx 109.54 / 2.045 \approx 53.57$

7. Answer: SE ≈ 9.78

- The margin of error equals $t^* \times \text{SE}$, so $\text{SE} = E / t^*$
- $\text{SE} = 20 / 2.045 \approx 9.78$

8. Answer: We are 95% confident that the true mean lifetime of AA batteries falls between 430 and 470 minutes.

- A confidence interval interpretation must reference the confidence level, the parameter (true population mean), and the interval bounds.
- Correct wording: 'We are 95% confident that the true mean lifetime of AA batteries is between 430 and 470 minutes.'



- Avoid saying 'there is a 95% probability the mean is in this interval' — the mean is a fixed (not random) value.

9. Answer: Point estimate = 16 oz; Margin of error = 0.4 oz; $t^* \approx 2.797$; $s \approx 0.715$ oz

- Point estimate = $(15.6 + 16.4) / 2 = 16$
- Margin of error = $16.4 - 16 = 0.4$
- Tail area = $(1 - 0.99) / 2 = 0.005$; $df = 25 - 1 = 24$
- Critical value: $t^*(0.005, 24) \approx 2.797$
- $s = E \times \sqrt{n} / t^* = 0.4 \times \sqrt{25} / 2.797 = 0.4 \times 5 / 2.797 = 2 / 2.797 \approx 0.715$

10. Answer: Point estimate = 60; $s \approx 19.44$; 99% CI $\approx (47.22, 72.78)$

- Point estimate = $(54.3 + 65.7) / 2 = 60$
 - 90% margin of error = $65.7 - 60 = 5.7$
 - Tail area for 90%: $(1 - 0.90) / 2 = 0.05$; $df = 19$; $t^*(0.05, 19) \approx 1.729$
 - Find s : $s = E \times \sqrt{n} / t^* = 5.7 \times \sqrt{20} / 1.729 = 5.7 \times 4.472 / 1.729 \approx 25.49 / 1.729 \approx 14.74$
 - Tail area for 99%: $(1 - 0.99) / 2 = 0.005$; $df = 19$; $t^*(0.005, 19) \approx 2.861$
 - 99% margin of error = $2.861 \times (14.74 / \sqrt{20}) = 2.861 \times 3.296 \approx 9.43$
 - 99% CI = $(60 - 9.43, 60 + 9.43) \approx (50.57, 69.43)$
 - Note: slight variation in final answer is acceptable depending on rounding of t^* and s .
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